

Terna Engineering College

Computer Engineering Department

Program: Sem VI

Course: Cloud Computing Lab(CSL605)

PART A

(PART A: TO BE REFERRED BY STUDENTS)

Experiment No.7

A.1 Aim:

To study Security as a Service on AWS

A.2 Prerequisite:

Knowledge of Access Control, Authentication and Authorization

A.3 Objective:

To understand the Security practices available in public cloud platforms and to study various Threat detection, Data protection and Infrastructure protection services in AWS.

A.4 Outcome: (LO 4)

After successful completion of this experiment student will be able to understand the Security practices available in public cloud platforms.

A.5 Theory:

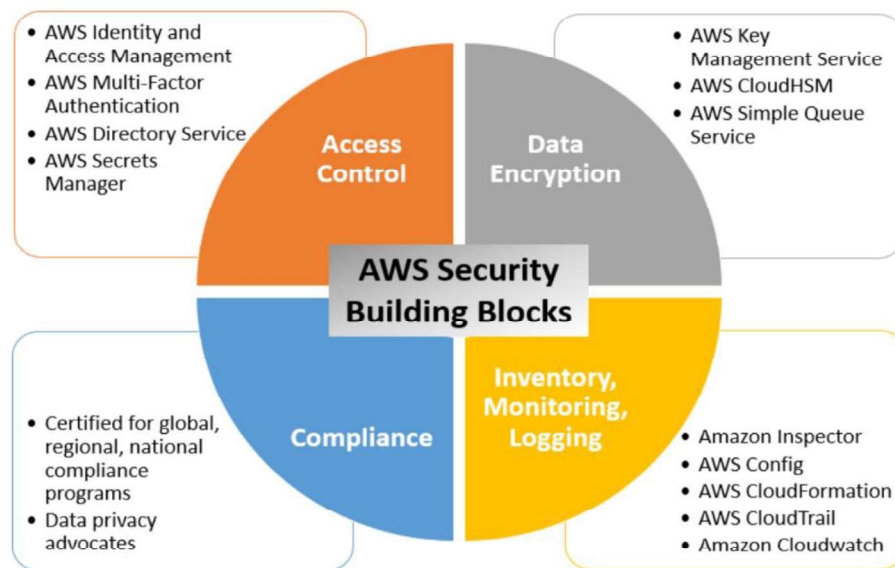
Amazon Web Services (AWS) enables organizations to build and scale applications quickly and securely. However, continuously adding new tools and services introduces new security challenges. According to reports, 70 percent of enterprise IT leaders are concerned about how secure they are in the cloud and 61 percent of small- to medium-sized businesses (SMBs) believe their cloud data is at risk.

AWS provides many different security tools to help customers keep their AWS accounts and applications secure.

What are the benefits of Security Services?

1. Keeps Data Safe: Infrastructure incorporates strong safeguards to help protect privacy. All data is processed in highly protected data centres.
2. Meets Compliance Requirements: Manages dozens of compliance programs in its infrastructure. Organizations meet compliance effortlessly
3. Saves Operational Cost: Operational cost reduces as organizations don't have to maintain on-premise facilities.

4. Scales Quickly: Security scales with the organization's usage of Amazon Web Services Cloud. The architecture is built to keep data secure, no matter the size of the enterprise.



AWS provides a number of security tools and services to help make your life easier when it comes to securing your cloud.

Data protection: AWS recognizes the importance of securing data and making sure it is not lost in transfer. Their services help you meet core security, confidentiality, and compliance requirements. Features include things encryption, data duplication and data monitoring. An example of a data protection service provided by AWS is Amazon Macie.

Identity & Access Management: AWS recognizes the need for managing Identities so they provide an extensive list of tools and services to help you manage identity in the cloud. Overall, the goal is to control the resources and actions identities can use and manipulate.

Infrastructure Protection: Infrastructure protection is a critical component of information security and helps ensure that everything within your workload is safe from vulnerability exploitation or unintended access. While infrastructure is largely managed by AWS itself, they also provide some additional resources for managing the security of configurable infrastructure, e.g. AWS WAF.

Threat Detection: When in the cloud, you need constant reassurance that your security posture is strong and you have all the right configurations in place to optimize security. AWS provides services that increase visibility into your deployment and operations and also monitor identity behavior to help detect threats. An example is Amazon GuardDuty.

PART B

(PART B: TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the ERP or emailed to the concerned lab in charge faculties at the end of the practical in case the there is no ERP access available)

Roll No. C54	Name: Quazi Md Moinuddin
Class : TE-C-COMPS	Batch : C3
Date of Experiment: 3-4-25	Date of Submission: 5-4-25
Grade :	

B.1 Question of Curiosity:

(To be answered by student based on the practical performed and learning/observations) Q.1.
What are AWS security monitoring and logging evaluation tools? Explain each.

Ans : AWS provides tools and features that enable you to see what's happening in your AWS environment. These include:

- With AWS CloudTrail, you can monitor your AWS deployments in the cloud by getting a history of AWS API calls for your account, including API calls made via the AWS Management Console, the AWS SDKs, the command line tools, and higher-level AWS services. You can also identify which users and accounts called AWS APIs for services that support CloudTrail, the source IP address the calls were made from, and when the calls occurred.
- Amazon CloudWatch, provides a reliable, scalable, and flexible monitoring solution that you can start using within minutes. You no longer need to set up, manage, and scale your own monitoring systems and infrastructure.
- Amazon GuardDuty, is a threat detection service that continuously monitors for malicious activity and unauthorized behavior to protect your AWS accounts and workloads. Amazon GuardDuty exposes notifications via Amazon CloudWatch so you can trigger an automated response or notify a human.
- Amazon Inspector, is an automated security assessment service that helps users identify security vulnerabilities and compliance issues in their AWS resources. Inspector assesses the security posture of EC2 instances, network configurations, and applications by running pre-defined security checks based on best practices and industry standards. It provides detailed findings and recommendations for remediation, allowing users to prioritize and address security issues proactively.

These tools and features give you the visibility you need to spot issues before they impact the business and allow you to improve security posture, and reduce the risk profile, of your environment.

Q.2 Explain cloud security.

Ans : Cloud security is the set of control-based security measures and technology protection, designed to protect online stored resources from leakage, theft, and data loss. Protection includes data from cloud infrastructure, applications, and threats. Security applications use a software the same as SaaS (Software as a Service) model.

How to manage security in the cloud :

Cloud service providers have many methods to protect the data.

Firewall is the central part of cloud architecture. The firewall protects the network and the perimeter of end-users. It also protects traffic between various apps stored in the cloud.

Access control protects data by allowing us to set access lists for various assets. For example, you can allow the application of specific employees while restricting others. It's a rule that employees can access the equipment that they require. We can keep essential documents which are stolen from malicious insiders or hackers to maintaining strict access control.

Data protection methods include Virtual Private Networks (VPN), encryption, or masking. It allows remote employees to connect the network. VPN accommodates the tablets and smartphone for remote access. Data masking maintains the data's integrity by keeping identifiable information private. A medical company share data with data masking without violating the HIPAA laws.

For example, we are putting intelligence information at risk in order of the importance of security. It helps to protect mission-critical assets from threats. Disaster recovery is vital for security because it helps to recover lost or stolen data.

Q.3 What are AWS Security use cases?

Ans : AWS (Amazon Web Services) offers a wide range of security use cases to address the diverse needs of organizations across different industries and sectors. These use cases encompass various aspects of security, including data protection, identity and access management, threat detection, and infrastructure security.

- One key use case is Identity and Access Management (IAM), which allows organizations to manage user identities and control access to AWS resources securely. IAM enables organizations to create and manage user accounts, assign permissions, and enforce multi-factor authentication to enhance security and compliance.
- Another important use case is Data Encryption, which involves encrypting data both in transit and at rest to protect it from unauthorized access. AWS provides encryption services such as AWS Key Management Service (KMS) and Amazon S3 encryption to help organizations safeguard sensitive data and maintain regulatory compliance.
- AWS Security Groups and Network ACLs (Access Control Lists) offer use cases for network security, allowing organizations to define and enforce firewall rules to control inbound and outbound traffic to their AWS resources. This helps organizations create secure network architectures and protect against unauthorized access and malicious attacks.
- Threat Detection and Monitoring is another critical use case provided by AWS through services such as Amazon GuardDuty and AWS CloudTrail. These services help organizations detect and respond to security threats in real-time by monitoring AWS environments for suspicious activities, unauthorized access attempts, and compliance violations.

- Compliance and Governance use cases involve implementing security best practices, regulatory compliance standards, and industry-specific requirements. AWS provides compliance services such as AWS Config, AWS Security Hub, and AWS Artifact to help organizations assess, monitor, and manage compliance across their AWS environments.

Q4 Define Amazon GuardDuty?

Ans : Amazon GuardDuty is a security monitoring service that analyzes and processes Foundational data sources, such as AWS CloudTrail management events, AWS CloudTrail event logs, VPC flow logs (from Amazon EC2 instances), and DNS logs. It also processes Features such as Kubernetes audit logs, RDS login activity, S3 logs, EBS volumes, Runtime monitoring, and Lambda network activity logs. It uses threat intelligence feeds, such as lists of malicious IP addresses and domains, and machine learning to identify unexpected, potentially unauthorized, and malicious activity within your AWS environment. This can include issues like escalation of privileges, use of exposed credentials, or communication with malicious IP addresses, domains, presence of malware on your Amazon EC2 instances and container workloads, or discovery of unusual patterns of login events on your database. For example, GuardDuty can detect compromised EC2 instances and container workloads serving malware, or mining bitcoin. It also monitors AWS account access behavior for signs of compromise, such as unauthorized infrastructure deployments, like instances deployed in a Region that hasn't been used before, or unusual API calls like a password policy change to reduce password strength.

GuardDuty informs you of the status of your AWS environment by producing security findings that you can view in the GuardDuty console or through Amazon EventBridge. GuardDuty also provides support for you to export your findings to an Amazon Simple Storage Service (S3) bucket, and integrate with other services such as AWS Security Hub and Detective.

Q5 What are the notable advantages of AWS security?

Ans : Security is our number one priority, and at AWS, we have comprehensive security capabilities to protect virtually any workload. Through AWS Security services, we provide the opportunities to protect your data, monitor security-related activity, and receive automated responses. This gives our customers the ability to increase their security posture in the cloud, deliver more agile IT, and lower costs.

The notable advantages of AWS security include:

1. **Robust Infrastructure:** AWS provides a highly secure and resilient infrastructure with multiple layers of protection, including physical security measures, network firewalls, and data encryption. This ensures the confidentiality, integrity, and availability of customer data.
2. **Compliance and Governance:** AWS offers a wide range of compliance certifications and adheres to strict regulatory standards, making it easier for organizations to comply with industry-specific requirements and government regulations.
3. **Identity and Access Management (IAM):** AWS IAM enables organizations to manage user access to AWS resources securely. With IAM, administrators can create and

manage user identities, control access permissions, and enforce multi-factor authentication (MFA) policies to enhance security.

4. **Data Encryption:** AWS provides encryption mechanisms to protect data both at rest and in transit. Services like AWS Key Management Service (KMS) allow organizations to create and manage encryption keys securely, while services like Amazon S3 and Amazon RDS offer built-in encryption options for data storage.
5. **Monitoring and Logging:** AWS offers a range of monitoring and logging services, such as Amazon CloudWatch and AWS CloudTrail, which provide real-time visibility into AWS resource utilization, application performance, and security-related events. These services help organizations detect and respond to security threats more effectively.
6. **Automated Security Controls:** AWS provides automated security controls and features, such as AWS Config and AWS Trusted Advisor, which help organizations identify security vulnerabilities, enforce compliance policies, and optimize their AWS infrastructure for security and cost-effectiveness.
7. **Scalability and Flexibility:** AWS security solutions are designed to scale with the needs of organizations, allowing them to adapt and grow their security posture as their AWS footprint expands. AWS offers a wide range of security services and features that can be tailored to meet the unique requirements of different workloads and applications.

B.3 Conclusion:

In conclusion, through this experiment, students have gained an understanding of the security practices available in public cloud platforms, particularly focusing on AWS. By studying various threat detection, data protection, and infrastructure protection services, students have learned the importance of implementing robust security measures in cloud environments. This knowledge equips them with the necessary skills to address security challenges effectively and ensure the safety of cloud-based applications and data.